

EfficientCaps: Robust Traffic Sign Recognition Under Real-World Degradation via Lightweight Capsule Networks

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INTRODUCTION

Traffic sign recognition (TSR) is **safety-critical in autonomous driving**. Standard benchmarks use clean images only. Real-world signs appear under:

- Snow & ice covering the sign surface
- Bullet holes, graffiti & stickers
- Night-time & motion blur
- Fading & physical damage from weathering

MOTIVATION & EXISTING CHALLENGES



Stop sign — bullet holes



Slippery road — snow cover

Can a model trained only on clean data generalize to real-world degraded signs through architecture and training strategy alone?

- Most models tested only on clean benchmarks
- Training on degraded images requires knowing which degradations to simulate
- Domain adaptation needs target domain data
- Hard visually similar classes (ice/snow vs slippery, curve L vs R)
- Traditional Capsule Networks have high computational cost

CONTRIBUTIONS

EfficientCaps: lightweight capsule network from scratch — 99.06% clean accuracy, no pre-training

Class-weighted margin loss + balanced sampler + hard-class augmentation for 10 difficult classes

inference pipeline: condition detection, TTA, disambiguation, reliability tiers

Single-step routing replacing iterative dynamic routing — lower cost, preserves equivariance

COMPUTATIONAL EFFICIENCY

14.20 M
Parameters

35 MFLOPs
FLOPs/img

166 FPS
Inference

90.7 MB
GPU Memory

~0.6 hr
Training (T4)

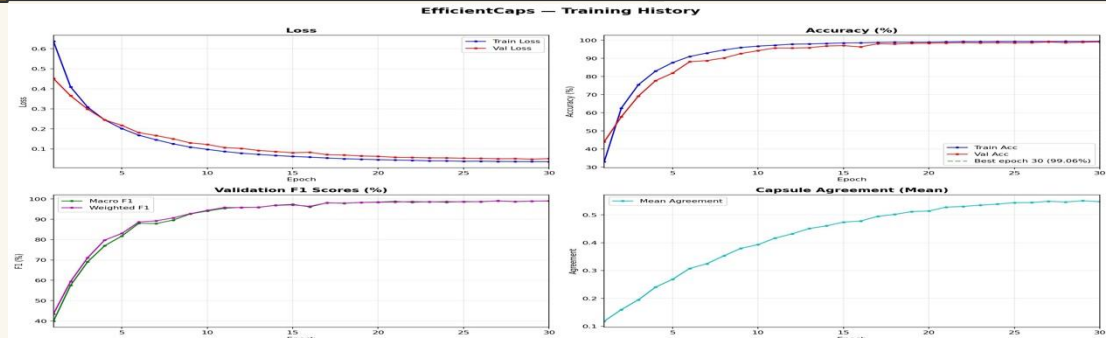
162.7 MB
Model size

ABLATION STUDY

WML	Hard Aug	Sampler	GTSRB	Degraded
—	—	—	98.3%	42.4%
✓	—	—	98.7%	45.5%
✓	✓	—	98.9%	48.5%
✓	✓	✓	99.06%	51.5% ★

Each component adds measurable gains on both clean and degraded data.

TRAINING HISTORY



METHODOLOGY — ARCHITECTURE



Backbone: Depthwise Separable Convolution Blocks
Routing: Single-Step Efficient Routing, no iterative EM
Decoder: Image Reconstruction

METHODOLOGY — TRAINING STRATEGY

$$\mathcal{L} = \sum_k \lambda_k [T_k \cdot \max(0, m^+ - \|v_k\|)^2 + \alpha \cdot (1 - T_k) \cdot \max(0, \|v_k\| - m^-)^2]$$

Ice/snow ×3.0

End no-pass ×3.0

Slippery ×2.5

Curve L ×2.0

Curve R ×2.0

Dbl curve ×2.0

Pedestrians ×2.0

Bumpy ×1.5

Caution ×1.5

Speed 20 ×1.5

Balanced sampler + hard-class augmentation

METHODOLOGY — INFERENCE PIPELINE

- Sign Detection** YOLOv8n detects & crops sign region — optional for pre-cropped inputs
- Condition Detection** Brightness, sharpness (Laplacian), saturation → night / snowy / blurry / normal
- Adaptive Enhancement** CLAHE (dark/snow), unsharp masking (blur), NL-Means (noise) — per condition
- Test-Time Augmentation** 4–8 augmented variants averaged; symmetric signs use fewer variants
- EfficientCaps Classification** Class Scores
- Disambiguation** Geometric checks for similar signs (symmetry analysis (23<->30), apex position (19<->20))
- Reliability Assessment** Prediction Confidence (HIGH / MEDIUM / LOW / UNKNOWN)
5 signals: confidence, capsule agreement, entropy, top-2 gap, per-class threshold

BASELINE COMPARISON (33 Real-World Images)

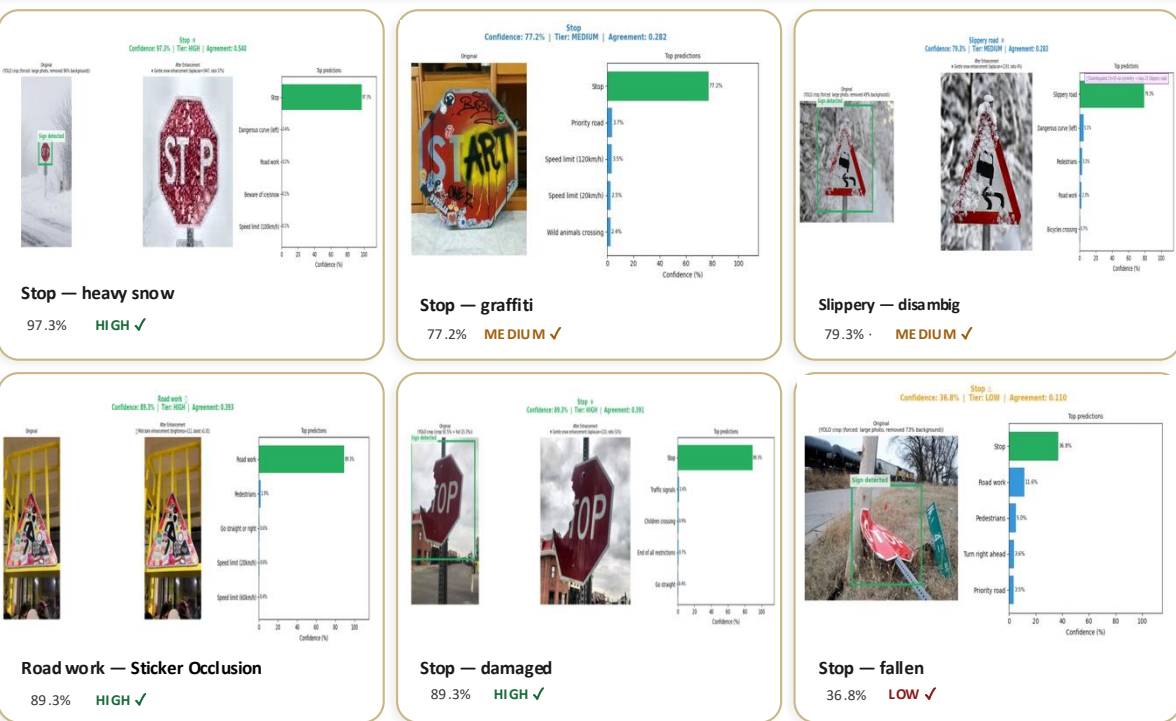
Model	GTSRB	Real-world
EfficientCaps V3 (Mine) ★	99.06%	51.5%
Cireşan et al. 2012	98.9%	45.5%
EfficientNet-B0	98.1%	45.5%
CSNN+CBAM	97.4%	48.5%
MobileNetV2	96.8%	48.5%

All models trained on clean GTSRB only. Evaluated on 33 real-world degraded images.

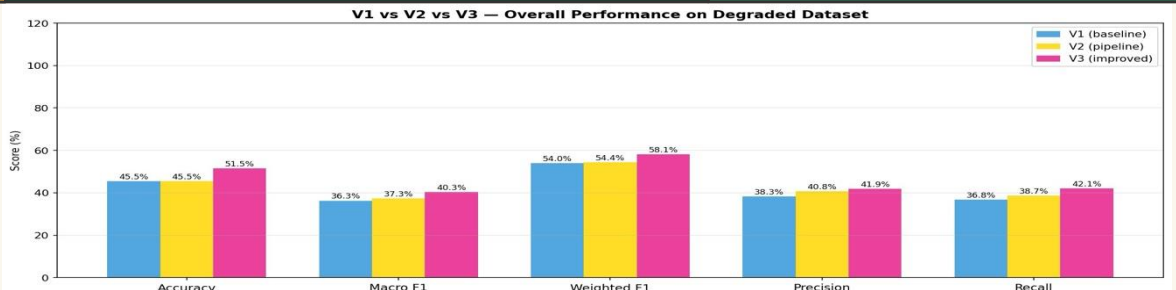
EXPERIMENT RESULTS

99.06% GTSRB Accuracy	96.5% Macro F1	Ep.30 Best Epoch
33/43 F1 ≥ 95%	10/43 F1 80–95%	0 F1 < 80%

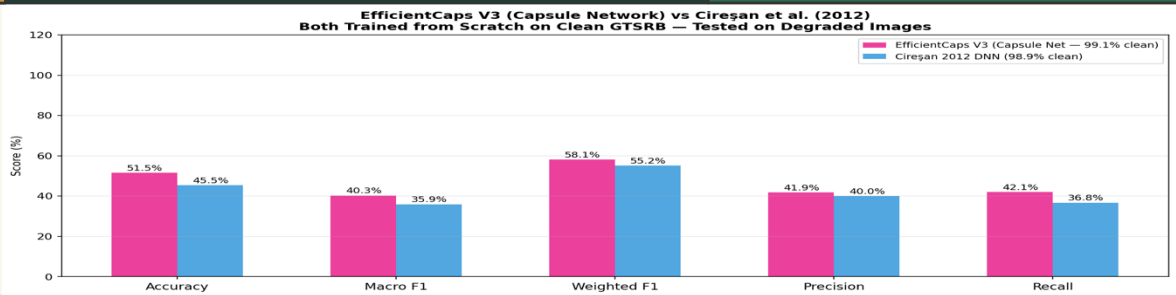
REAL-WORLD PREDICTIONS



V1 vs V2 vs V3



EfficientCaps V3 vs Cireşan (2012)



EfficientCaps V3 vs CSNN+CBAM

